

Lizbeth Alexandra Torres Flores

Diplôme d'ingénieur généraliste (in progress) · Electronic Engineering

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I am pursuing the Diplôme d'ingénieur généraliste at IMT Atlantique through a double-degree agreement with UNI, where I graduated in Electronic Engineering. I design and validate embedded systems, IoT prototypes, FPGA logic and computer-vision methods. My work includes VIS-SWIR research at Institut Pascal, electrical-room lighting and UPS engineering at ABB Peru, and Python development at the University of Alberta.

Technical focus: Python, C/C++, VHDL; ESP32, STM32, EFR32MG24 and FPGA; OpenCV and PyTorch; IoT communications; DIALux and UPS engineering.

Education

IMT Atlantique AUG 2025-PRESENT
Diplôme d'ingénieur généraliste (Bac+5) · Double degree Brest / Rennes, France

Selected for the IMT Atlantique-UNI double-degree programme. In year two, I followed IoT and 5G for Industry 4.0. In year three, I am studying Embedded and Heterogeneous Systems, with practical work on Zynq-7000/ZedBoard, ARM, FPGA logic, AXI, Vivado/Vitis and Linux.

Universidad Nacional de Ingeniería (UNI) 2020-2025
Electronic Engineering Lima, Peru

At UNI, I studied electronic circuits, programming, digital systems and control. I applied that training by designing, programming and testing physical prototypes; my mathematics and engineering projects are presented separately in the Projects section.

Professional and research experience

Institut Pascal APR-AUG 2026
Computer Vision Research Intern Université Clermont Auvergne · Clermont-Ferrand, France

At Institut Pascal, I researched the alignment of road images captured by visible (VIS) and short-wave infrared (SWIR) cameras. Differences in contrast and texture make reliable cross-modal matches difficult.

- Co-designed and implemented the RASMD benchmark protocol: four weather conditions, both registration directions, 100 image pairs per task and four complementary metrics.
- Implemented point matching and homography estimation. Patch-MAGSAC++ combines sparse patch-wise SuperPoint/LightGlue matches with MAGSAC++ for a global homography.
- Developed and evaluated RoMa-MIND, combining dense RoMa matches with MIND-based multimodal refinement.
- Automated data preparation, registration and evaluation in Python, OpenCV, PyTorch and Makefile; built a Streamlit interface to inspect images and methods.

Deliverables: Reproducible protocol, both method implementations, bidirectional experiments and comparison interface; Weather-, direction- and metric-specific analysis with visual overlays and discrepancies.

Result: The two methods improved on different baselines depending on the task. For sunny SWIR to VIS, RoMa-MIND achieved 3.43 px reprojection error against 14.81 px for ORB/RANSAC. Patch-MAGSAC++ performed particularly well in several VIS to SWIR tasks. No method led across every weather condition, direction and metric; a global homography also leaves local depth-related misalignment.

Tools: Python · OpenCV · PyTorch · SuperPoint · LightGlue · MAGSAC++ · RoMa · MIND · Makefile · Streamlit

ABB Peru DEC 2024-AUG 2025
Engineering intern · Process Industries / Process Automation Process Industries Division · Lima, Peru

At ABB Peru, I worked on electrical-room lighting and UPS engineering for mining and industrial projects in Peru and Colombia. Each room needed its own lighting study and electrical specifications.

- Modelled electrical rooms in 3D in DIALux and selected luminaire types and placement for indoor, outdoor and emergency lighting.
- Checked required illuminance levels for each space and adjusted the design to project and country requirements.
- Sized UPS batteries and chargers using IEEE 1115 and IEEE 485. Migrated the first part of a previously manual Excel process to Python.
- The Python tool takes input data and generates calculations, plots and an Excel file, avoiding repeated manual work in that stage.
- Prepared cable and material lists for electrical rooms with Autodesk Navisworks.

Deliverables: Room-specific DIALux lighting studies and designs for several mining sites; UPS and battery calculations, plots and Excel output from Python; cable and material lists.

Result: The studies supported review of illuminance and luminaire choice for each room. The Python workflow turned the first UPS-sizing stage into a repeatable process that generates calculations, plots and Excel from input data.

Tools: DIALux · 3D modelling · Python · Excel · Autodesk Navisworks · IEEE 1115 / IEEE 485 · UPS

University of Alberta FEB-MAY 2024
International Research Intern Edmonton, Canada

During the UARE placement at the University of Alberta, I worked on two separate Python engineering tasks: exam-figure generation and migration of an existing urban-solar analysis.

- Developed parameterized 2D and 3D figures in Python and Jupyter; changing dimensions or inputs automatically generates the corresponding exam images.
- Migrated existing MATLAB analysis code for an urban solar-adoption study to Python.
- Implemented only the first part of the migrated tool in MecSimCalc for interactive use.

Deliverables: Parameterized figure generator for assessment materials; First stage of the Python analysis made interactive in MecSimCalc.

Result: Exam-figure generation was automated. The solar analysis was migrated to Python, and its first stage was available interactively in MecSimCalc; the study yielded preliminary results.

Tools: Python · Jupyter · MATLAB · 2D/3D visualization · MecSimCalc

MMJ Smart Electronics DEC 2023-NOV 2024

Embedded Systems Intern Lima, Peru

MMJ Smart Electronics built custom electronics and embedded solutions for different client companies. My internship combined prototyping, hardware/software integration, testing and documentation.

- Worked on several client prototypes using ESP32, STM32, sensors, GPS and communications, from assembly and integration to functional checks.
- Fabricated, soldered and tested PCBs and wrote manuals for boards, equipment and prototypes.
- Tested a locally developed PLC prototype and contributed to its user guide; also worked on measurement devices and electrical cabinets.
- As part of the internship, prepared IoT course material and supported practical electronics and hardware workshops.

Deliverables: Electronic prototypes for different clients, assembled and checked PCBs, PLC tests and technical documentation; Electronics and IoT teaching material and workshops delivered within the internship.

Result: The prototypes, tests and manuals supported custom deliveries and demonstrations. Results from different clients are kept distinct.

Tools: ESP32 · STM32 · PCB · soldering · GPS · sensors · PLC · electrical cabinets · IoT

Smart City Lab, UNI APR 2023-FEB 2024

Research Intern Lima, Peru

At UNI's Smart City Lab, I worked on physical robotics and computer vision prototypes, each with a separate design and test sequence.

- Developed the complete documented vision and control chain of a line-following robot with Raspberry Pi and OpenCV and tested it on a physical course.
- Designed and simulated a 16-servo biped in SolidWorks Motion and programmed its gait.
- Contributed to a 3D-printed EMG-controlled transradial prosthesis using MATLAB and Python for signal processing.

Deliverables: Line-following robot, humanoid gait demonstration and prosthesis prototype, each with its own visual material; Technical presentation of the prosthesis at the second International Engineering for Health Congress.

Result: The three prototypes were demonstrated in the lab. The College of Engineers of Peru certificate confirms my participation as a presenter on the prosthesis.

Tools: Raspberry Pi · OpenCV · Python · SolidWorks Motion · Arduino · EMG · MATLAB · 3D printing

Engineering projects

VIS-SWIR image registration benchmark COMPUTER VISION · 2026

- Visible and SWIR cameras capture complementary information, yet their road images do not align automatically. Contrast, weather and scene depth complicate comparison and fusion.
- I co-designed and implemented the bidirectional RASMD benchmark, worked on Patch-MAGSAC++ and RoMa-MIND, automated experiments and built the Streamlit results interface.
- Result: Eight tasks cover four weather conditions and both directions, with 100 pairs per task and GOA, EDGA, MIND-SSD and reprojection error. For sunny SWIR to VIS, RoMa-MIND reached 3.43 px versus ORB/RANSAC's 14.81 px; Patch-MAGSAC++ led other VIS to SWIR comparisons.
- Two registration strategies: Patch-MAGSAC++ finds sparse SuperPoint/LightGlue matches in image patches and estimates a global homography with MAGSAC++. RoMa-MIND starts with dense RoMa matches and refines cross-modal alignment using MIND descriptors.
- Evaluation protocol: Sunny, cloudy, rainy and snowy scenes are compared in both directions. Gradient agreement (GOA), edge agreement (EDGA), multimodal MIND-SSD and geometric reprojection error expose different strengths that one score would conceal.
- Scope and limits: The interface shows overlays and per-task results. No method dominates every case. A global homography can leave local errors around vertical objects and surfaces at different depths.
- Tools: Python, OpenCV, PyTorch, Makefile, Streamlit

CEREBRO · sensor-network security SECURE IOT · OCT 2026-ONGOING

- Security can change latency, packet delivery, timing drift and energy use in a wireless sensor network. The project will compare configurations under equivalent conditions.
- After common Contiki-NG and Cooja onboarding, I will work with a teammate on nRF52840 boards: build and flash, establish initial unsecured communication, then study secured tests and available cryptographic acceleration.
- Result: The project is in progress; there are no final measured metrics or conclusions. Simulation and the comparison interface were initially assigned to the other pair.
- My planned role: Set up the physical nRF52840 testbed, verify unsecured communication first and document configuration, problems and results. Then assess Contiki-NG security mechanisms on hardware.
- Evaluation plan: The team plans to compare an unsecured configuration with two secured ones, measuring latency, packet delivery, timing drift and energy under varied network conditions. These are planned measures, not achieved findings.
- Tools: Contiki-NG, nRF52840, IEEE 802.15.4, TSCH / RPL

Eco4Impact · shared TCO core INDUSTRIAL SOFTWARE · SEP 2025-MAR 2026

- Vectura System needed to compare transport choices by initial investment, operating costs and end-of-life resale value. Eco4Impact and EcoBoatTwin needed common functions without duplicating the model.
- I structured truck and country information in JSON, defined calculation inputs and worked on residual-value formulas, coordinated with the team's common CAPEX, OPEX and RV architecture.
- Result: At completion, the two teams had a shared TCO calculation core with modules and tests for trucks and boats. My individual contribution is distinguished from the overall team output.
- Model inputs: JSON organizes vehicle characteristics, energy types, country, costs, taxes and technical parameters. Consistent fields let scenarios and modules use comparable inputs.

- Residual value: End-of-period value depends on age, use, vehicle condition, technology evolution and external factors. This component prevents a TCO comparison based only on purchase price and energy cost.
- Team result: The modular Python architecture combines CAPEX, OPEX and residual value. Final documentation includes truck and boat examples; public examples omit private client parameters.
- Tools: Python, JSON, CosApp, TCO, CAPEX / OPEX / RV

Voice commands on EFR32MG24 EMBEDDED AI · 2026

- The goal was to recognize commands on a microcontroller without external processing and verify a physical output.
- I prepared MFCC features in Edge Impulse, integrated the model and audio pipeline in C/C++ with TensorFlow Lite Micro and Micrium OS, flashed the board and connected inference to GPIO.
- Result: The board demonstration showed the complete voice-to-LED path. No unmeasured accuracy, latency or power figures are claimed.
- Processing chain: Audio becomes MFCC features, the model classifies them locally and the application turns the recognized command into a GPIO action.
- Hardware validation: The application was built, flashed to EFR32MG24 and observed controlling the LED in response to voice commands.
- Tools: EFR32MG24, C/C++, MFCC, TensorFlow Lite Micro, Micrium OS, Edge Impulse

8-bit VHDL processor on FPGA DIGITAL DESIGN · 2026

- The IMT Atlantique lab supplied the architecture and some starter files. I completed and extended this educational CPU and documented how to check it.
- I completed the control unit and FSMs, wrote a readable RTL datapath alternative to the supplied netlist, added programs and testbenches, and organized Vivado variants and Nexys A7 documentation.
- Result: The repository has VHDL sources, scripts and reproducible tests. Board photographs document the lab setup; the full automated suite has not yet been successfully rerun and is documented separately.
- Architecture: An accumulator and unified 64 × 8-bit memory support NOR, ADD, STA and JCC. Standard and optimized FSMs allow cycle comparison; the RTL datapath exposes data flow.
- Tests and attribution: Testbenches cover instructions, control signals, memory, ADD and NOR cases. The repository distinguishes lab starter files from my extensions and records the published test status.
- Tools: VHDL, Vivado, FSM, FPGA, Nexys A7
- GitHub: https://github.com/lizbethtorresflores/cpu8bits-vhdl

FPGA traffic-light controller DIGITAL CONTROL · 2026

- A road junction needs a safe light sequence with defined timing. This project realizes the controller as observable digital logic on FPGA.
- I designed the VHDL state machine and timing counters, checked transitions in simulation, then deployed it on Nexys4 and wired LED outputs to an MDF junction model.
- Result: Waveforms and the physical demonstration showed the light sequence.
- Controller design: States define each junction phase and counters trigger transitions. Keeping control separate from timing made sequence review possible before physical wiring.
- Verification: Waveforms showed state and output changes; the lights were then checked on Nexys4 and the physical model.
- Tools: VHDL, FSM, Nexys4

Portable vital-sign monitor IOT PROTOTYPE · 2024-2025

- My UNI final project explored remote acquisition and transmission of physiological readings as an IoT engineering prototype, without clinical validation.
- I integrated MAX30102, MLX90614 and MPX5050DP sensors on ESP32, programmed acquisition and transmission over LoRaWAN and GSM/SIM800L, and built a Ubidots dashboard.
- Result: Tests showed readings arriving and displaying in real time. The technical contribution is sensor, connectivity and dashboard integration.
- Data path: ESP32 gathers pulse/SpO2, temperature and pressure readings. Communication modules send data to Ubidots for time-series inspection.
- Verified scope: Tests covered prototype transmission and visualization. The device is not described as certified medical equipment.
- Tools: ESP32, Sensors, LoRaWAN, GSM, Ubidots

EMG-controlled bionic prosthesis ROBOTICS · 2023-2024

- The Smart City Lab explored a 3D-printed transradial prosthesis driven by electromyographic signals.
- I contributed to electronic and mechanical integration, prototype fabrication, EMG acquisition and signal processing in MATLAB and Python to connect muscle activity with movement.
- Result: The prototype was demonstrated in the lab, and I presented its design and implementation at the College of Engineers of Peru's second International Engineering for Health Congress.
- From muscle to movement: EMG signals were acquired and processed before becoming control commands. The work combined this chain with 3D-printed parts and electronic assembly.
- Public presentation: The College of Engineers certificate confirms my participation as a presenter on the design and implementation of the prosthesis.
- Tools: EMG, 3D printing, MATLAB, Python

16-servo bipedal humanoid ROBOTICS · 2023-2024

- Bipedal walking required coordinated joint sequences suited to the robot's geometry and 16 servomotors.
- I designed the 3D structure in SolidWorks, simulated movement in SolidWorks Motion, calculated joint angles and programmed servo sequences in Arduino IDE.
- Result: The gait was demonstrated on a physical Smart City Lab prototype.
- Design and simulation: The mechanical model helped review joints and movement ranges before programming. Simulation structured leg and balance sequences.
- Implementation: Calculated angles became commands for 16 servos on the physical robot. No unmeasured speed or stability figures are claimed.
- Tools: SolidWorks Motion, Arduino, Servomotors

Autonomous line-following robot EMBEDDED VISION · 2023

- The vehicle needed to detect the line in its own camera frames and correct its direction as it moved.
- I developed the complete vision and control chain: HSV color segmentation, region of interest, line detection and grouping, lateral offset and steering calculation, and motor actuation from Raspberry Pi.
- Result: The robot followed a controlled test course. The report includes images of detection and the vehicle on the route.
- Line detection: OpenCV converts camera frames to HSV to isolate the reference line. A region of interest restricts analysis; edge and line detection estimate the visible path.
- From image to motors: I computed offset from the image center and steering angle from detected lines. Raspberry Pi sent commands to the motor stage.
- Test scope: Testing took place on a known, controlled track. The result establishes line following in that environment.
- Tools: Raspberry Pi, OpenCV, Python, HSV, motor control

Graph-colouring timetable generator MATHEMATICS & SOFTWARE · 2020-II

- At UNI, incompatible classes or lab sections forced students to rebuild schedules. Some prepared 10-12 options manually.
- I represented courses and sections as vertices and timing or selection conflicts as edges. Graph colouring and search generated compatible combinations and exported alternative schedules.
- Result: The project won second place in the UNI Faculty of Sciences 2020-II mathematics-project competition. The report notes that the search can grow exponentially.
- Conflict model: Each course option has a day and time slot. Overlapping or incompatible options are connected in a graph; a valid selection avoids conflicts.
- Generating alternatives: A backtracking search explores combinations and exports compatible schedules for comparison, replacing manual reconstruction. Its complexity limits large cases.
- Tools: graphs, colouring, C++, backtracking

Least-squares fitting and applications APPLIED MATHEMATICS · 2020-I

- Observations seldom lie exactly on a theoretical curve. The project studied parameter selection that minimizes residuals and the limits of subsequent predictions.
- I studied linear fitting through normal equations and nonlinear fitting with Gauss-Newton; the team implemented MATLAB examples and examined a logistic model on COVID-19 data.
- Result: Presented at the UNI Faculty of Sciences 2020-I mathematics-project fair. The data application illustrates a mathematical method, not a validated epidemiological forecast.
- Methods: The report moves from line and curve fitting to nonlinear models, where Gauss-Newton iteratively adjusts parameters to reduce residuals.
- COVID-19 example: A logistic curve was fitted to cumulative data to discuss possible predictions and the limits of extrapolation.
- Tools: MATLAB, least squares, Gauss-Newton, logistic model

Technical skills

Programming and technical software	Python for automation, analysis, data models and interactive tools; C/C++ for embedded systems; MATLAB for migration and signal processing; VHDL for digital design. Git, Linux, Jupyter, Makefile and JSON. **Applied in:** *Institut Pascal · ABB · Alberta · Eco4Impact · CPU*
Embedded and real-time systems	Integration and tests on ESP32, STM32, EFR32MG24, Raspberry Pi and Arduino; GPIO, sensors, audio acquisition and Micrium OS. Zynq-7000, Vitis and embedded Linux are current lab work; Zephyr is introductory knowledge. **Applied in:** *MMJ · voice control · IoT monitor · IMT labs*
FPGA and digital design	VHDL, state machines, counters, simulation, synthesis, bitstreams and tests on Nexys A7/Nexys4 with Vivado. Lab starter modules are distinguished from my CPU extensions. **Applied in:** *8-bit CPU · traffic-light controller*
Computer vision and multimodal registration	OpenCV and PyTorch; image processing, matching, homographies and metric-based comparison. SuperPoint, LightGlue, MAGSAC++, RoMa and MIND in the VIS-SWIR benchmark. **Applied in:** *Institut Pascal · line-following robot*
On-device applied AI	MFCC feature preparation with Edge Impulse and TensorFlow Lite Micro integration for local inference under Micrium OS, validated on EFR32MG24. **Applied in:** *Voice-controlled LED*
IoT, sensors and communication	Sensor and GPS integration, LoRaWAN, GSM/SIM800L, MQTT and Ubidots. Introductory I2C, SPI, Modbus and RS485; Contiki-NG/nRF52840 security is ongoing work. **Applied in:** *IoT monitor · MMJ · CEREBRO*
Electrical engineering and automation	DIALux electrical-room lighting, IEEE 1115/485 UPS sizing, Excel-to-Python automation and Navisworks cable/material lists; PLC and PCB tests. **Applied in:** *ABB Peru · MMJ*
Mechanical design and prototyping	SolidWorks and SolidWorks Motion for mechanisms; 3D printing, CAD, assembly and tests. AutoCAD, Inventor and Revit appear as additional training. **Applied in:** *Humanoid · prosthesis · UNI/CTIC training*
Research and validation methods	Comparative protocols, data preparation, metrics, limitations, technical documentation and communication in international teams. **Applied in:** *VIS-SWIR · Alberta · UNI projects*

Leadership and presentations

IEEE EMBS UNI Student Chapter 2023
HR director and deputy events director from June to December; volunteer in technical projects throughout the year. Coordinated more than six technical activities and gave an EEG/EMG presentation.

UNI student representation 2022-2024
Elected student representative on the Faculty Council (2022-23); served on academic, research, accreditation, quality and enrolment committees, as well as faculty hiring and promotion committees. Joined the Electronic Engineering Department Council in 2024.

IEEE WIE UNI · Girls in ICT Day 2024
Speaker encouraging girls and young women to consider careers in information and communications technology.

Airbus Fly Your Ideas · Team CÓNDOR 2026
Qualified for round two of the Airbus Fly Your Ideas Challenge with team CÓNDOR.

Aspire Leaders Program International programme
I joined a leadership, communication and social-impact programme for first-generation students.

Parliament of Women · Peru Civic participation
I took part in legislative simulations and public-speaking workshops on women's representation in STEM and civic life.

ABB Peru internal committees 2024-2025
I participated in diversity and gender-equity and social-responsibility initiatives within ABB Peru.

Applications and perspectives of EEG and EMG signals
Presented how biomedical signals can be acquired and used in engineering applications for the IEEE EMBS UNI Student Chapter.

Girls in ICT Day
Spoke at an event encouraging young women to explore technology and engineering careers.

3D-printed bionic prosthesis for transradial amputations
Presented the design and implementation of the prototype at the 2nd International Congress of Engineering for Health, organized by the professional engineers' association in Lima.

Selected additional training

SENATI · industrial electronics: Power Electronics (18 h), Basic Industrial Instrumentation (18 h), Industrial Sensors and Actuators (16 h), and Fundamentals of Refrigeration and Air Conditioning (12 h).

UNI / CTIC · engineering software: Intermediate Python; AutoCAD 1 and 2; Autodesk Inventor; BIM with Revit; cloud computing across AWS, Azure and Google Cloud.

MMJ & I&T Bionics: PCB Design and Fabrication, 24 hours (2024).

ABB · industrial practice: Lean Six Sigma White and Yellow Belts, HSE Guiding Principles, BeSecure onboarding and control-panel solutions.

Platzi · software and teamwork: Electronic Circuit Design, PIC Microcontrollers in C, Software Engineering Fundamentals, Scrum, Agile Delivery Management, Agile Team Management, Project Management Fundamentals and development-environment setup.

Languages

Spanish: native · French: B2 · English: B2